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United States Patent	12408001
Kind Code	B2
Date of Patent	September 02, 2025
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Rendering of audio signals using virtualized reverberation

Abstract

A computer-implemented method for processing audio. The method includes obtaining a binaural room impulse response (BRIR) of an acoustic space, receiving an input audio signal, separating the input audio signal into low-frequency components and high-frequency components, and dividing the BRIR of the acoustic space into a first portion that occurs before a first time and a second portion that occurs after the first time. The method further includes generating a first component of an output audio signal based on the high-frequency components of the input audio signal and the first portion of the BRIR, generating a second component of the output audio signal based on the high-frequency components of the input audio signal and the second portion of the BRIR, generating a third component of the output audio signal based on the low-frequency components of the input audio signal and the BRIR, and outputting the output audio signal.

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Appl. No.: 18/336267

Filed: June 16, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20240422501 A1	Dec. 19, 2024

Publication Classification

Int. Cl.: H04S7/00 (20060101); H04S3/00 (20060101)

U.S. Cl.:

CPC **H04S7/306** (20130101); **H04S3/008** (20130101); **H04S7/304** (20130101); **H04S7/307** (20130101); H04S2400/01 (20130101); H04S2400/11 (20130101); H04S2400/13 (20130101); H04S2400/15 (20130101); H04S2420/11 (20130101)

Field of Classification Search

USPC: None

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Background/Summary

BACKGROUND

Field of the Various Embodiments

(1) The various embodiments relate generally to audio processing, and more specifically, to techniques for rendering audio signals using virtualized reverberation.

DESCRIPTION OF THE RELATED ART

(2) Reverberation refers to the persistence of sound in an enclosed or semi-enclosed space after the sound has been produced by a source. When sound is produced in an acoustic space, such as a room, concert hall, movie theater, etc., the sound propagates from one or more sources as a series of sound waves. The propagating sound waves travel through the air and reflect off boundaries of the acoustic space, such as walls, ceilings, and floors, and/or objects included in the acoustic space. As the reflected sound waves continue to bounce off the boundary surfaces and other objects within the acoustic space, these reflections blend together and gradually decay over time as the sound energy is absorbed by surfaces and/or objects in the acoustic space.

(3) To create a more immersive audio experience when using headphones or ear buds, the above-described reverberation effects can be artificially added to audio signals using digital signal processing techniques. For example, in music production, a sound engineer can render audio signals with reverberation to simulate the effects of listening to music in a particular acoustic environment, such as a concert hall or an auditorium. As another example, in film or game production, audio signals can be rendered with reverberation to recreate the sounds of acoustic environments depicted on screen, such as caves, corridors, or specific outdoor spaces. In other examples, such as augmented or virtual reality applications, audio signals rendered with reverberation allow users to perceive an enhanced sense of presence and immersion with the augmented or virtual world.

(4) Various digital signal processing techniques and algorithms, which offer tradeoffs between computational efficiency and accuracy, can be used for rendering reverberation to simulate the effects of an acoustic space during playback of an audio signal. Some virtualized reverberation algorithms render reverberation in an audio signal by convolving the audio signal with a room impulse response (RIR). An RIR is the time-domain acoustic transfer function between a sound source and a receiver in a given acoustic space (e.g., a concert hall, an auditorium, etc.). The RIR of an acoustic space can be measured, for example using loudspeakers and a microphone, or simulated using acoustic modelling software. One drawback to convolving an audio signal with an RIR is that when using high order processing techniques, convolving the long reverberation tails included in an RIR with the audio signal is computationally very expensive and time consuming.

(5) Other techniques divide the RIR into a first window of direct, or early arriving, sound and a second window of reflected, or later arriving, sound (e.g., the reverberation tail). The window of direct sound in the RIR is convolved with the audio signal using high order, computationally intensive processing techniques. The window of reflected sound in the RIR is then separately convolved with the audio signal using lower order, less computationally intensive processing techniques. These additional techniques, however, typically use an RIR derived from a model of an acoustic space because it is difficult and computationally expensive to obtain windows of direct sound and reflected sound from a measured RIR of an acoustic space. When compared to a measured RIR of an acoustic space, an RIR derived from the model of an acoustic space can be difficult to obtain and/or generate and often fails to capture all of the reverberation effects of an acoustic space.

(6) As the foregoing illustrates, what is needed are more effective techniques for rendering audio signals with reverberation.

SUMMARY

(7) Various embodiments of the present disclosure set forth a computer-implemented method for processing audio. The method includes obtaining a binaural room impulse response (BRIR) of an acoustic space, receiving an input audio signal, separating the input audio signal into low-frequency components and high-frequency components, and dividing the BRIR of the acoustic space into a first portion that occurs before a first time and a second portion that occurs after the first time. The method further includes generating a first component of an output audio signal based on the high-frequency components of the input audio signal and the first portion of the BRIR, generating a second component of the output audio signal based on the high-frequency components

of the input audio signal and the second portion of the BRIR, generating a third component of the output audio signal based on the low-frequency components of the input audio signal and the BRIR, and outputting the output audio signal

(8) Further embodiments provide, among other things, one or more non-transitory computer-readable media and systems configured to implement the method set forth above.

(9) At least one technical advantage of the disclosed techniques relative to the prior art is that, with the disclosed techniques, the measured RIR of an acoustic space can be used to add reverberation to an audio signal with a lower computational cost. Accordingly, with the disclosed techniques, relatively modest processing power can be used to render reverberation for a large number of sound sources and acoustic spaces based on measured room impulse responses, which sound more natural than room impulse responses derived from models of acoustic spaces. These technical advantages provide one or more technological improvements over prior art approaches.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) So that the manner in which the above recited features of the various embodiments can be understood in detail, a more particular description of the inventive concepts, briefly summarized above, may be had by reference to various embodiments, some of which are illustrated in the appended drawings. It is to be noted, however, that the appended drawings illustrate only typical embodiments of the inventive concepts and are therefore not to be considered limiting of scope in any way, and that there are other equally effective embodiments.

(2) FIG. 1 is a block diagram of a computing device, according to one or more aspects of the various embodiments;

(3) FIG. 2 is a block diagram of a system for measuring a binaural room impulse response of an acoustic space, according to one or more aspects of the various embodiments;

(4) FIG. 3 illustrates the mid-to-high-frequency components of a measured BRIR of an acoustic space, according to one or more aspects of the various embodiments;

(5) FIG. 4 illustrates the low-frequency components of a measured BRIR of an acoustic space, according to one or more aspects of the various embodiments;

(6) FIG. 5 illustrates a block diagram of a technique for processing an audio signal with a virtualized reverberation algorithm based on measured binaural room impulse responses, according to one or more aspects of the various embodiments;

(7) FIG. 6 illustrates a block diagram of a convolution process included in the technique for processing audio illustrated in FIG. 5, according to one or more aspects of the various embodiments;

(8) FIG. 7 illustrates a block diagram of a technique for processing an audio signal with a virtualized reverberation algorithm based on measured binaural room impulse responses, according to one or more aspects of the various embodiments; and

(9) FIG. 8 is a flow chart of method steps for processing an audio signal, according to one or more aspects of the various embodiments.

DETAILED DESCRIPTION

(10) In the following description, numerous specific details are set forth to provide a more thorough understanding of the various embodiments. However, it will be apparent to one of skill in the art that the inventive concepts may be practiced without one or more of these specific details.

(11) FIG. 1 is a block diagram of a computing device **100**, according to various embodiments. As will be described in more detail below, computing device **100** can be used to render audio signals using virtualized reverberation techniques. Computing device **100** can be implemented as any type of computing device, such as a desktop computer, a laptop computer, a mobile device, a server, or a

smart audio output device, suitable for practicing the various embodiments. As shown, computing device **100** includes, without limitation, one or more processing units **102**, network interface **104**, input/output (I/O) devices interface **106**, input device(s) **108**, output device(s) **110**, system storage **112**, and system memory **114**. Computing device **100** further includes an interconnect **116** that is configured to facilitate transmission of data, such as programming instructions and application data, between processing unit(s) **102**, network interface **104**, I/O devices interface **106**, system storage **112**, and system memory **114**.

(12) Processing unit(s) **102** can be any technically feasible processing device configured to process data and execute program instructions. For example, processing unit(s) **102** could include one or more central processing units (CPUs), digital signal processors (DSPs), graphics processing units (GPUs), application-specific integrated circuits (ASICs), field-programmable gate arrays (FPGAs), microprocessors, microcontrollers, other types of processing units, and/or a combination of different processing units. Processing unit(s) **102** is configured to retrieve and execute programming instructions, such as binaural room impulse response (BRIR) measurement application **118** and virtualized reverberation application **120**, stored in system memory **114**. Similarly, processing unit(s) **102** is configured to store application data (e.g., software libraries) and retrieve application data from system storage **112** and/or system memory **114**.

(13) Computing device **100** can connect with other devices and/or networks via network interface **104**. In some embodiments, network interface **104** is hardware, software, or a combination of hardware and software, that is configured to connect to and interface with one or more devices and/or networks. In some embodiments, network interface **104** facilitates communication with other devices or systems via one or more standard and/or proprietary protocols (e.g., Bluetooth, a proprietary protocol associated with a specific manufacturer, etc.)

(14) I/O devices interface **106** is configured to receive input data from input device(s) **108** and transmit the input data to processing unit(s) **102** via interconnect **116**. For example, input device(s) **108** can include one or more buttons, a keyboard, a mouse, a graphical user interface, a touchscreen display, a microphone, and/or other input devices. I/O devices interface **106** is further configured to receive output data from processing unit(s) **102** via interconnect **116** and transmit the output data to output device(s) **110**. For example, output device(s) **110** can include one or more of a display device, a touchscreen display, a graphical user interface, a loudspeaker, and/or other output devices.

(15) System storage **112** can include non-volatile storage for applications, software modules, and data, and can include fixed or removable disk drives, flash memory devices, and CD-ROM, DVD-ROM, Blu-Ray, HD-DVD, or other magnetic, optical, solid state storage devices, and/or the like. System storage **112** can be fully or partially located in a remote storage system, referred to herein as “the cloud,” and accessed through network connections via network interface **104**. System storage **112** is configured to store non-volatile data such as files (e.g., audio files, video files, subtitles, application files, software libraries, etc.). In one example, system storage **112** stores binaural room impulse response (BRIR) data **122** associated with the measured BRIRs of one or more acoustic spaces. A binaural room impulse response refers to the acoustic impulse response between a sound source and a listener's ears in an acoustic space. Thus, each BRIR includes a left ear component of the BRIR and a right ear component of the BRIR.

(16) System memory **114** can include a random access memory (RAM) module, a flash memory unit, or any other type of memory unit or combination thereof. One or more of processing unit(s) **102**, network interface **104**, and I/O devices interface **106**, are configured to read data from and write data to system memory **114**. System memory **114** includes various software programs and modules (e.g., an operating system, one or more applications) that can be executed by processing unit(s) **102** and application data (e.g., data loaded from system storage **112**) associated with said software programs. For example, as will be described in more detail below, system memory **114** includes BRIR measurement application **118** and virtualized reverberation application **120**.

(17) BRIR measurement application **118** can be executed by processing unit(s) **102** to measure the

BRIR of an acoustic space and store the measured BRIR of the acoustic space as BRIR data **122** in system storage **112**. FIG. **2** is a block diagram of an example system **200** for measuring the binaural room impulse response of an acoustic space **202**. Acoustic space **202** can be any type of room or listening environment for which it is desired to simulate a listening experience with reverberation effects. For example, acoustic space **202** can be a concert hall, an auditorium, a theater, a stadium, or some other type of listening environment. As shown, system **200** includes computing device **100** executing BRIR measurement application **118**, a loudspeaker **204**, and an omni-directional microphone **206**. Although loudspeaker **204** is illustrated as a single loudspeaker, in some embodiments, system **200** includes multiple loudspeakers distributed around acoustic space **202**. In some embodiments, loudspeaker **204** includes multiple speaker drivers. For example, loudspeaker **204** includes one or more tweeters, mid-range drivers, and woofers.

(18) In operation, BRIR measurement application **118** controls loudspeaker **204** to emit audio signals within acoustic space **202**. Omni-directional microphone **206** records the emitted audio signals propagating through acoustic space **202** and BRIR measurement application **118** processes the recorded audio signals to determine the BRIR of acoustic space **202**. BRIR measurement application **118** can use one or more known techniques, such as the maximum-length sequence (MLS) technique, the inverse repeated sequence (IRS) technique, the time-stretched pulses technique, or the logarithmic sinesweep technique, to measure and determine the BRIR of acoustic space **202**. BRIR measurement application **118** then stores the measured BRIR of acoustic space **202** as BRIR data **122** in system storage **112**. As described above, the BRIR of acoustic space **202** is the acoustic impulse response between a sound source and a listener's ears in acoustic space **202**. Thus, the BRIR of acoustic space **202** includes a left ear component of the BRIR and a right ear component of the BRIR.

(19) Referring back to FIG. **1**, when executed by processing unit(s) **102**, virtualized reverberation application **120** renders audio signals with reverberation based on a measured BRIR of an acoustic space. For example, virtualized reverberation application **120** renders an audio signal with reverberation based on the BRIR of acoustic space **202** that was measured by BRIR measurement application **118** and stored as BRIR data **122** in system storage **112**. Rendering an audio signal with reverberation based on a measured BRIR of an acoustic space includes adding reverberation to the audio signal to simulate the effects of listening to the playback of audio in the acoustic space. In some embodiments, when rendering an audio signal with reverberation based on a measured BRIR, virtualized reverberation application **120** filters, or separates, the measured BRIR into the mid-to-high-frequency components of the measured BRIR and the low-frequency components of the measured BRIR. As used herein, the term mid-to-high-frequency can be used interchangeably with the term high-frequency.

(20) FIG. **3** illustrates the mid-to-high-frequency components **300** of a measured BRIR of an acoustic space, according to one or more aspects of the various embodiments. For example, the FIG. **3** illustrates the mid-to-high-frequency components **300** of the BRIR of acoustic space **202** measured by BRIR measurement application **118**. The mid-to-high-frequency components **300** of the measured BRIR include the components of the measured BRIR that have frequencies greater than a crossover frequency. In some embodiments, the value of the crossover frequency is determined empirically. In some embodiments, the value of the crossover frequency is determined based on one or more characteristics of acoustic space **202**. In one non-limiting example, the crossover frequency is 900 Hz. In other examples, the value of the crossover frequency is less than 900 Hz or greater than 900 Hz. Hereinafter, the mid-to-high-frequency components **300** of the measured BRIR of acoustic space **202** can also be referred to simply as the high-frequency components of the measured BRIR of acoustic space **202**.

(21) As shown in FIG. **3**, the mid-to-high-frequency components of the measured BRIR include a direct sound portion **302** and a reflected sound portion **304**. The direct sound portion **302** includes the portion of sound emitted by loudspeaker **204** that is received directly by omni-directional

microphone **206** when BRIR measurement application **118** measures the BRIR of acoustic space **202**. That is, the direct sound portion **302** is the portion of sound emitted by loudspeaker **204** that arrives directly at omni-directional microphone **206** without being reflected off one or more surfaces (e.g., walls, floor, ceiling, structures, etc.) of acoustic space **202**. The time at which the direct sound portion **302** arrives at omni-directional microphone **206** is indicative of the time at which sound emitted by a source, such as loudspeaker **204**, in acoustic space **202** arrives directly at a listener's ears.

(22) In contrast, the reflected sound portion **304** includes the portions of sound emitted by loudspeaker **204** that are reflected off one or more surfaces of acoustic space **202** before arriving at omni-directional microphone **206** when BRIR measurement application **118** measures the BRIR of acoustic space **202**. The time at which the reflected sound portion **304** arrives at omni-directional microphone **206** is indicative of the time at which sound emitted by a source, such as loudspeaker **204**, that reflects off one or more surfaces in acoustic space **202** arrives at a listener's car drums. The reflected sound portion **304** can be referred to as the “reverberation tail” of the measured BRIR of acoustic space **202**. As shown in FIG. 3, the peak amplitude of the direct sound portion **302** is generally greater than the peak amplitudes of the reflected sound portion **304**. This occurs because some of the energy of a sound wave is absorbed by a surface of acoustic space **202** when the sound wave is reflected off the surface of acoustic space **202** and because the sound wave attenuates as the sound wave travels a further distance.

(23) As further shown in FIG. 3, the direct sound portion **302** arrives at omni-directional microphone **206** before the reflected sound portion **304** arrives at omni-directional microphone **206**. This occurs because the path of a sound wave that reflects off one or more surfaces of acoustic space **202** before arriving at omni-directional microphone **206** is longer than the path of sound that travels directly from the source (e.g., loudspeaker **204**) to omni-directional microphone **206**. Therefore, when rendering an audio signal with reverberation, virtualized reverberation application **120** can divide the mid-to-high-frequency components **300** of the measured BRIR of acoustic space **202** into the direct sound portion **302** and the reflected sound portion **304** temporally. That is, virtualized reverberation application **120** can divide the mid-to-high-frequency components **300** of the measured BRIR into a first time window **306** and second time window **308** to separate the direct sound portion **302** from the reflected sound portion **304**. When compared to other techniques for processing a BRIR of an acoustic space, dividing a measured BRIR of an acoustic space temporally into a first time window and a second time window is computationally inexpensive.

(24) In the illustrated example, the first time window **306** includes the time before time **T1** and the second time window **308** includes the time after time **T1**. The value of time **T1** is selected to be a time by which the direct sound portion **302** has already arrived at omni-directional microphone **206** but also a time by which the reflected sound portion **304** has not yet arrived at omni-directional microphone **206**. The value of time **T1** is dependent on one or more characteristics of acoustic space **202**, as the amount of time taken for the direct and/or reflected sound portions **302**, **304** to arrive at omni-directional microphone **206** can vary among different acoustic spaces. In some embodiments, virtualized reverberation application **120** determines the value of time **T1** automatically by analyzing the mid-to-high-frequency components **300** of the measured BRIR. In some embodiments, the value of time **T1** is determined empirically and/or defined manually by a user.

(25) In one non-limiting example, the value of time **T1** is 2 ms. In this example, the direct sound portion **302** included in the mid-to-high-frequency components **300** of the measured BRIR arrives at omni-directional microphone **206** within 2 ms of being emitted by loudspeaker **204**. Continuing with this example, the reflected sound portion **304** included in the mid-to-high-frequency components **300** of the measured BRIR arrives at omni-directional microphone **206** more than 2 ms after being emitted by loudspeaker **204**. In some examples, the value of time **T1** is less than 2 ms. In other examples, the value of time **T1** is greater than 2 ms.

(26) In the illustrated example of FIG. 3, virtualized reverberation application **120** implements a hard windowing technique to divide the mid-to-high-frequency components **300** of the measured BRIR into a first time window **306** that occurs before time **T1** and second time window **308** that occurs after time **T1** to separate the direct sound portion **302** from the reflected sound portion **304**. When virtualized reverberation application **120** implements the hard windowing technique, in which there is no overlap between the first and second time windows **306**, **308**, to separate the direct sound portion **302** from the reflected sound portion **304**, there are no negative effects that might typically be attributed to abruptly cutting off an impulse response. In some embodiments, virtualized reverberation application **120** implements a soft windowing technique, in which there is some overlap between the first and second time windows **306**, **308**, to separate the direct sound portion **302** from the reflected sound portion **304**.

(27) As will be described in more detail below, dividing the mid-to-high-frequency components **300** of the measured BRIR into first and second time windows **306**, **308** allows virtualized reverberation application **120** to render the mid-to-high-frequency components of an audio signal with the direct sound portion **302** using a precise, yet computationally more expensive, processing technique and separately render the mid-to-high-frequency components of an audio signal with reflected sound portion **304** using a less precise, yet computationally more efficient, processing technique. For example, virtualized reverberation application **120** implements the reverberant loudspeaker (RVL) method to discretely render the mid-to-high-frequency components of an audio signal with the direct sound portion **302** included in the mid-to-high-frequency components **300** of the measured BRIR. As another example, virtualized reverberation application **120** implements a low order processing technique in which the mid-to-high-frequency components of an audio signal are encoded and decoded before rendering the mid-to-high-frequency components of the audio signal with the reflected sound portion **304** included in the mid-to-high-frequency components of the measured BRIR.

(28) FIG. 4 illustrates the low-frequency components of a measured BRIR of an acoustic space, according to one or more aspects of the various embodiments. For example, the FIG. 4 illustrates the low-frequency components **400** of the BRIR of acoustic space **202** measured by BRIR measurement application **118**. The low-frequency components **400** of the measured BRIR include the components of the measured BRIR that have frequencies less than a crossover frequency. In some embodiments, the value of the crossover frequency is determined empirically. In some embodiments, the value of the crossover frequency is determined based on one or more characteristics of acoustic space **202**. In one non-limiting example, the crossover frequency is 900 Hz. In other examples, the value of the crossover frequency is less than 900 Hz or greater than 900 Hz.

(29) Similar to the mid-to-high-frequency components **300** of the measured BRIR, the low-frequency components **400** of the measured BRIR also include a direct sound portion **402** and a reflected sound portion **404**. However, unlike the mid-to-high-frequency components **300** of the measured BRIR, the direct and reflected sound portion **402**, **404** included in the low-frequency components **400** of the measured BRIR cannot easily be separated temporally. For example, as shown in FIG. 4, there is overlap between the times at which direct sound portion **402** arrives at omni-directional microphone **206** and the times at which reflected sound portion **404** arrives at omni-directional microphone **206**. This overlap in arrival times can be attributed to the longer wavelengths of low-frequency sound waves when compared to the wavelengths of mid-to-high-frequency sound waves. Thus, even if virtualized reverberation application **120** divides the low-frequency components **400** of the measured BRIR into a first time window **406** that occurs before a time **T1** and a second time window **408** that occurs after time **T1**, such windowing does not separate the direct sound portion **402** from the reflected sound portion **404**. In some instances, attempting to separate the direct sound portion **402** from the reflected sound portion **404** when rendering the low-frequency components of an audio signal can introduce undesired windowing

effects, which are not an issue at higher frequencies. Accordingly, virtualized reverberation application **120** does not separate the direct sound portion **402** from the reflected sound portion **404** when rendering the low-frequency components of an audio signal.

(30) FIG. 5 illustrates a block diagram **500** of a technique for processing an audio signal with a virtualized reverberation algorithm based on measured binaural room impulse responses, according to one or more aspects of the various embodiments. The technique can be implemented, for example, by virtualized reverberation application **120** executing on processing unit(s) **102** of computing device **100**. In some embodiments, virtualized reverberation application **120** implements the processing technique illustrated in FIG. 5 using the BRIR of acoustic space **202** that was measured by BRIR measurement application **118** and stored in system storage **112** as BRIR data **122**.

(31) As shown, virtualized reverberation application **120** receives an input audio signal to render with reverberation. In the illustrated example of FIG. 5, the input audio signal is monoaural audio signal that includes a single audio channel. However, in some embodiments, the input audio signal includes more than one channel of audio. For example, the input audio signal can include two or more audio channels.

(32) Virtualized reverberation application **120** applies a filtering process **502** to the input audio signal to separate the low-frequency components of the input audio signal from the mid-to-high-frequency components of the input audio signal. For example, the filtering process **502** includes applying a low-pass filter to the input audio signal to obtain the low-frequency components of the input audio signal. The low-frequency components of the input audio signal include the components of the input audio signal that have frequencies less than a crossover frequency. Filtering process **502** further includes applying a high-pass filter to the input audio signal to obtain the mid-to-high-frequency components of the input audio signal. The mid-to-high-frequency components of the input audio signal include the components of the input audio signal that have frequencies greater than the crossover frequency. As described above, the value of the crossover frequency can be determined empirically and/or based on one or more characteristics of an acoustic space. In one non-limiting example, the crossover frequency is 900 Hz. Hereinafter, the mid-to-high-frequency components of an audio signal, such as the input audio signal, can also simply be referred to as the high-frequency components of an audio signal.

(33) After separating the low-frequency and mid-to-high-frequency components of the input audio signal, virtualized reverberation application **120** uses a first processing technique **504** to add reverberation to the low-frequency components of the input audio signal. Furthermore, virtualized reverberation application **120** uses second and third processing techniques **506**, **508** to add reverberation to the mid-to-high-frequency components of the input audio signal. When compared to third processing technique **508**, which can be implemented as a discrete RVL method for adding reverberation to the discrete mid-to-high-frequency components of the input audio signal, first and second processing techniques **504**, **506** are less precise but more computationally efficient. That is, first and second processing techniques **504**, **506** consume less computing resources when implemented by virtualized reverberation application **120** than third processing technique **508**.

(34) As shown, first processing technique **504** includes an encoding and decoding process **510** in which virtualized reverberation application **120** encodes low-frequency components of the single audio channel of the input audio signal into a plurality of encoded channels of low-frequency audio. While implementing encoding and decoding process **510**, virtualized reverberation application **120** further decodes the plurality of encoded channels of low-frequency audio into a plurality of decoded channels of low-frequency audio. In some instances, the plurality of decoded channels of low-frequency audio correspond to the cardinal directions (e.g., top, bottom, left, right, front, back) from which sound arrives at the ears of a listener. Hereinafter, the decoded channels of low-frequency audio output by encoding and decoding process **510** can be referred to as converted channels of low-frequency audio and/or converted low-frequency audio channels.

(35) In the illustrated example, virtualized reverberation application **120** uses a low-order (e.g., first order) Ambisonics encoder-decoder to encode the single audio channel of the input audio signal into four low-frequency Ambisonics-encoded audio channels (e.g., Ambisonics B format) and decode the four low-frequency Ambisonics-encoded audio channels into six low-frequency audio channels. Each of the six low-frequency decoded, or converted, audio channels corresponds to a respective direction (e.g., top, bottom, left, right, front, back) from which sound arrives at the ears of a listener. Persons skilled in the art will understand that, in some embodiments, virtualized reverberation application **120** uses other types of low-order encoder-decoders to implement encoding and decoding process **510**. Moreover, persons skilled in the art will understand that in some embodiments, encoding and decoding process **510** encodes and decodes (e.g., converts) the single audio channel of the input audio signal into fewer than or more than six channels of low-frequency audio.

(36) First processing technique **504** further includes a gain reduction process **512** in which virtualized reverberation application **120** reduces the gain of the converted channels of low-frequency audio. In the illustrated example of FIG. 5, gain reduction process **512** reduces the gain of the six low-frequency converted audio channels. Virtualized reverberation application **120** applies gain reduction process **512** to the plurality of converted low-frequency audio channels to prevent rendering an audio signal with reverberation in which the mid-to-high-frequency components of the rendered audio signal are overpowered by the low frequency components. In one non-limiting example, gain reduction process **512** reduces the respective gains of the plurality of converted low-frequency audio channels to values between -8 dB and -15 dB. However, persons skilled in the art will understand that in some embodiments, gain reduction process **512** reduces the respective gains of the plurality of converted low-frequency audio channels to other values.

(37) Virtualized reverberation application **120** then applies a low-frequency convolution process **514** to the plurality of converted low-frequency audio channels to generate left and right low-frequency audio channels that include reverberation effects. Low-frequency convolution process **514** includes convolving the plurality of converted low-frequency audio channels with low-frequency components of a measured BRIR of an acoustic space. As described above, because it is difficult to divide the low-frequency components of a measured BRIR into direct and reflected sound portions in a computationally efficient manner (e.g., temporally), low-frequency convolution process **514** convolves the plurality of converted low-frequency audio channels with both the direct and reflected sound portions included in the low-frequency components of the measured BRIR. For example, when implementing low-frequency convolution process **514**, virtualized reverberation application **120** convolves the plurality of converted low-frequency audio channels with the direct and reflected sound portions **402**, **404** included in the low-frequency components **400** of the measured BRIR of acoustic space **202**.

(38) FIG. 6 illustrates a block diagram **600** of a convolution process that can be used to implement low-frequency convolution process **514**. With respect to FIG. 6, during low-frequency convolution process **514**, virtualized reverberation application **120** convolves each of the plurality of converted low-frequency audio channels with respective low-frequency left ear components of the measured BRIR. When a respective converted low-frequency audio channel is convolved with a corresponding low-frequency left ear component of the measured BRIR, the result is a rendered low-frequency audio channel that includes reverberation effects. After convolving each converted low-frequency audio channel with a corresponding low-frequency left ear component of the measured BRIR, virtualized reverberation application **120** sums the resultant rendered low-frequency audio channels that into a single rendered low-frequency left ear audio channel that includes reverberation effects. In some embodiments, each of the low-frequency left ear components of the measured BRIR corresponds to a respective direction (e.g., top, bottom, left, right, front, back) from which sound arrives at the left ear of a listener.

(39) Similarly, during low-frequency convolution process **514**, virtualized reverberation application

120 convolves each of the plurality of converted low-frequency audio channels with respective low-frequency right car components of the measured BRIR. When a respective converted low-frequency audio channel is convolved with a corresponding low-frequency right car component of the measured BRIR, the result is a rendered low-frequency audio channel that includes reverberation effects. After convolving each converted low-frequency audio channel with a corresponding low-frequency right car component of the measured BRIR, virtualized reverberation application **120** sums the resultant rendered low-frequency audio channels into a single rendered low-frequency right car audio channel that includes reverberation effects. In some embodiments, each of the low-frequency right car components of the measured BRIR corresponds to a respective direction (e.g., top, bottom, left, right, front, back) from which sound arrives at the right car of a listener.

(40) As described above, after virtualized reverberation application **120** applies low-frequency convolution process **514** to the plurality of converted low-frequency audio channels, the result is a rendered low-frequency left car audio channel and a rendered low-frequency right car audio channel. First processing technique **504** further includes a headphone equalization process **516** in which virtualized reverberation application **120** equalizes the rendered low-frequency left ear audio channel and the rendered low-frequency right car audio channel. Headphone equalization process **516** implements one or more known equalization techniques to adjust the respective volume levels of frequency components included in the rendered low-frequency left and right car audio channels.

(41) Referring back to FIG. 5, virtualized reverberation application **120** also uses second and third processing techniques **506**, **508** to add reverberation to the mid-to-high-frequency components of the input audio signal. As shown in FIG. 5, second processing technique **506** includes an encoding and decoding process **518** in which virtualized reverberation application **120** encodes the mid-to-high-frequency components of the single audio channel of the input audio signal into a plurality of encoded channels of mid-to-high-frequency audio. While implementing encoding and decoding process **518**, virtualized reverberation application **120** further decodes the plurality of encoded channels of mid-to-high-frequency audio into a plurality of decoded channels of mid-to-high-frequency audio. In some instances, the plurality of decoded channels of mid-to-high-frequency audio correspond to the cardinal directions (e.g., top, bottom, left, right, front, back) from which sound arrives at the cars of a listener. Hereinafter, the decoded channels of mid-to-high-frequency audio output by encoding and decoding process **510** can be referred to as converted channels of mid-to-high-frequency audio and/or converted mid-to-high-frequency audio channels.

(42) In the illustrated example, virtualized reverberation application **120** uses a low-order (e.g., first order) Ambisonics encoder-decoder to encode the single audio channel of the input audio signal into four mid-to-high-frequency Ambisonics-encoded audio channels (e.g., Ambisonics B format) and decode the four mid-to-high-frequency Ambisonics-encoded audio channels into six mid-to-high-frequency audio channels. Each of the six mid-to-high-frequency decoded, or converted, audio channels corresponds to a respective direction (e.g., top, bottom, left, right, front, back) from which sound arrives at the cars of a listener. However, persons skilled in the art will understand that, in some embodiments, virtualized reverberation application **120** uses other types of low-order encoder-decoders to implement encoding and decoding process **518**. Moreover, persons skilled in the art will understand that in some embodiments, encoding and decoding process **518** encodes and decodes (e.g., converts) the single audio channel of the input audio signal into fewer than or more than six channels of mid-to-high-frequency audio.

(43) Second processing technique **508** further includes a gain reduction process **520** in which virtualized reverberation application **120** reduces the gain of the plurality of converted channels of mid-to-high-frequency audio. In the illustrated example of FIG. 5, gain reduction process **520** reduces the gain of the six mid-to-high-frequency converted audio channels. Virtualized reverberation application **120** applies gain reduction process **520** to the plurality of converted low-frequency audio channels to prevent rendering an audio signal with reverberation in which the mid-

to-high-frequency components rendered with direct sound portions of a measured BRIR are overpowered by mid-to-high-frequency components rendered with reflected sound portions of a measured BRIR. In one non-limiting example, gain reduction process **520** reduces the respective gains of the plurality of converted mid-to-high-frequency audio channels to values between -8 dB and -15B. However, persons skilled in the art will understand that in some embodiments, gain reduction process **520** reduces the respective gains of the plurality of converted mid-to-high-frequency audio channels to other values.

(44) Virtualized reverberation application **120** then applies a reflected convolution process **522** to the plurality of converted mid-to-high-frequency audio channels to generate left and right audio channels that include reverberation effects of reflected sound arriving at a listener's ears. Reflected convolution process **522** includes convolving the plurality of converted mid-to-high-frequency audio channels with only the reflected sound portions included in the mid-to-high-frequency components of a measured BRIR of an acoustic space. For example, when implementing reflected convolution process **522**, virtualized reverberation application **120** convolves the plurality of converted mid-to-high-frequency audio channels with only the reflected sound portion **304** included in the mid-to-high-frequency components **300** of the measured BRIR of acoustic space **202**. As described above with respect to FIG. 3, the reflected sound portion **304** included in the mid-to-high-frequency components **300** of a measured BRIR can be separated from the direct sound portion **302** included in the mid-to-high-frequency components **300** temporally.

(45) The block diagram **600** of the convolution process illustrated in FIG. 6 can also be used to implement reflected convolution process **522**. With respect to FIG. 6, during reflected convolution process **522**, virtualized reverberation application **120** convolves each of the plurality of converted mid-to-high-frequency audio channels with respective reflected sound portions included in the mid-to-high-frequency left ear components of the measured BRIR. When a respective converted mid-to-high-frequency audio channel is convolved with a corresponding reflected sound portion included in the mid-to-high-frequency left ear component of the measured BRIR, the result is a rendered mid-to-high-frequency audio channel that includes reverberation effects associated with the reflected sound portion of a measured BRIR. After convolving each converted mid-to-high-frequency audio channel with a corresponding reflected sound portion included in the mid-to-high-frequency left ear component of the measured BRIR, virtualized reverberation application **120** sums the resultant rendered mid-to-high-frequency audio channels into a single rendered mid-to-high-frequency left ear audio channel that includes reverberation effects associated with the reflected sound portion of the measured BRIR. In some embodiments, each of the reflected sound portions included in the mid-to-high-frequency left ear components of the measured BRIR corresponds to a respective direction (e.g., top, bottom, left, right, front, back) from which reflected sound arrives at the left ear of a listener.

(46) Similarly, during reflected convolution process **522**, virtualized reverberation application **120** convolves each of the plurality of converted mid-to-high-frequency audio channels with respective reflected sound portions included in the mid-to-high-frequency right ear components of the measured BRIR. When a respective converted mid-to-high-frequency audio channel is convolved with a corresponding reflected sound portion included in the mid-to-high-frequency right ear component of the measured BRIR, the result is a rendered mid-to-high-frequency audio channel that includes reverberation effects associated with the reflected sound portion of a measured BRIR. After convolving each converted mid-to-high-frequency audio channel with a corresponding reflected sound portion included in the mid-to-high-frequency right ear component of the measured BRIR, virtualized reverberation application **120** sums the resultant rendered mid-to-high-frequency audio channels into a single rendered mid-to-high-frequency right ear audio channel that includes reverberation effects associated with the reflected sound portion of the measured BRIR. In some embodiments, each of the reflected sound portions included in the mid-to-high-frequency right ear components of the measured BRIR corresponds to a respective direction (e.g., top, bottom, left,

right, front, back) from which reflected sound arrives at the right car of a listener.

(47) As described above, after virtualized reverberation application **120** applies reflected convolution process **522** to the plurality of converted mid-to-high-frequency audio channels, the result is a rendered mid-to-high-frequency left car audio channel that includes reverberation effects associated with the reflected sound portion of a measured BRIR and a rendered mid-to-high-frequency right car audio channel that includes reverberation effects associated with the reflected sound portion of a measured BRIR. Second processing technique **506** further includes a headphone equalization process **524** in which virtualized reverberation application **120** equalizes the above-described rendered mid-to-high-frequency left and right car audio channels that include reverberation effects associated with the reflected sound portion of a measured BRIR. Headphone equalization process **524**, which is similar to headphone equalization process **516**, implements one or more known equalization techniques to adjust the respective volume levels of frequency components included in the rendered mid-to-high-frequency left and right car audio channels.

(48) With respect to third processing technique **508**, virtualized reverberation application **120** does not encode and decode, or convert, the mid-to-high-frequency components of the single audio channel included in the input audio signal before implementing direct convolution process **526**. Rather, virtualized reverberation application **120** renders the discrete mid-to-high-frequency components of the single audio channel included in input audio signal with reverberation using a high-resolution, direct convolution process **526**. In the illustrated example of FIG. 5, third processing technique **508** is implemented using the RVL method for adding reverberation to an audio signal.

(49) When applying direct convolution process **526** to the discrete mid-to-high-frequency components of single audio channel included in the input audio signal, virtualized reverberation application **120** convolves the discrete mid-to-high-frequency audio channel with only the direct sound portion included in the mid-to-high-frequency components of a measured BRIR of an acoustic space. For example, when implementing direct convolution process **526**, virtualized reverberation application **120** convolves the discrete mid-to-high-frequency audio channel with the only the direct sound portion **302** included in the mid-to-high-frequency components **300** of the measured BRIR of acoustic space **202**. As described above with respect to FIG. 3, the direct sound portion **302** included in the mid-to-high-frequency components **300** of a measured BRIR can be separated from the reflected sound portion **304** included in the mid-to-high-frequency components **300** temporally.

(50) In one example, when implementing direct convolution process **526**, virtualized reverberation application **120** convolves the discrete mid-to-high-frequency audio channel with the direct sound portion included in the mid-to-high-frequency left ear components of the measured BRIR. The direct sound portion included in the mid-to-high-frequency left ear components of the measured BRIR corresponds to the direction at which sound emitted by a source arrives directly at the left ear of a listener. Similarly, when implementing direct convolution process **526**, virtualized reverberation application **120** also convolves the discrete mid-to-high-frequency audio channel with the direct sound portion included in the mid-to-high-frequency right car components of the measured BRIR. The direct sound portion included in the mid-to-high-frequency right car components of the measured BRIR corresponds to the direction at which sound emitted by a source arrives directly at the right car of a listener.

(51) The result of direct convolution process **526** is a rendered mid-to-high-frequency left car audio channel that includes reverberation effects associated with the direct sound portion of a measured BRIR and a rendered mid-to-high-frequency right car audio channel that includes reverberation effects associated with the direct sound portion of a measured BRIR. Using the high-resolution, direct convolution process **525** to convolve the discrete mid-to-high-frequency audio channel with the direct sound portion included in the mid-to-high-frequency components of the measured BRIR helps offset potential performance losses attributed to using low-order encoding and decoding in

the second processing technique **506**. As described above, in some examples, a low-order Ambisonics encoder-decoder is used to implement the low-order encoding and decoding in the second processing technique **506**.

(52) Third processing technique **508** further includes a headphone equalization process **528** in which virtualized reverberation application **120** equalizes the rendered mid-to-high-frequency left and right car audio channels that include reverberation effects associated with the direct sound portion of a measured BRIR. Headphone equalization process **528**, which is similar to headphone equalization processes **516** and **524**, implements one or more known equalization techniques to adjust the respective volume levels of frequency components included in the rendered mid-to-high-frequency left and right car audio channels.

(53) Virtualized reverberation application **120** then implements a summation process **530** to combine the rendered left car audio channels generated using first, second, third processing techniques **504**, **506**, and **508** and to combine the rendered right car audio channels generated using first, second, third processing techniques **504**, **506**, and **508** into a rendered stereo audio signal that includes a left audio channel and a right audio channel. For example, when implementing summation process **530**, virtualized reverberation application **120** sums the rendered low-frequency left car audio channel generated using first processing technique **504**, the rendered mid-to-high-frequency left car audio channel generated using second processing technique **506**, and the rendered mid-to-high-frequency left ear audio channel generated using third processing technique **508** to obtain a single left car audio channel that includes reverberation effects. Similarly, when implementing summation process **530**, virtualized reverberation application **120** sums the rendered low-frequency right car audio channel generated using first processing technique **504**, the rendered mid-to-high-frequency right car audio channel generated using second processing technique **506**, and the rendered mid-to-high-frequency right car audio channel generated using third processing technique **508** to obtain a single right car audio channel that includes reverberation effects. Virtualized reverberation application **120** then outputs a rendered stereo audio signal that includes the single left car audio channel and the single right car audio channel for playback by headphones or other loudspeaker arrangement capable of playing back stereo audio.

(54) In the illustrated example of FIG. 5, the input audio signal is a monaural audio signal that includes a single audio channel. However, in some instances, the input audio signal includes more than one channel of audio. For example, the input audio signal can include two or more audio channels. FIG. 7 illustrates a block diagram **700** of a technique for processing an audio signal of N channels with a virtualized reverberation algorithm based on measured binaural room impulse responses, according to one or more aspects of the various embodiments.

(55) As shown, the processing technique illustrated in FIG. 7 is very similar in operation to the processing technique illustrated in FIG. 5. However, whereas the processing technique illustrated in FIG. 5 includes a single filtering process **502** for separating the low-frequency components of the mono input audio signal from the mid-to-high-frequency components of the mono input audio signal, the processing technique illustrated in FIG. 7 includes n filtering processes **702-1-702-N** for separating the low-frequency components of the N audio channels included in the input audio signal from the mid-to-high-frequency components of the N audio channels included in the input audio signal.

(56) For example, virtualized reverberation application **120** implements first filtering process **702-1** to separate the low-frequency components of a first audio channel included in the input audio signal from the mid-to-high-frequency components of the first audio channel included the input audio signal. For example, the first filtering process **702-1** includes applying a low-pass filter to the first audio channel included in the input audio signal to obtain the low-frequency components of the first audio channel. The low-frequency components of the first audio channel included in the input audio signal include the components of the first audio channel that have frequencies less than a crossover frequency. First filtering process **702-1** further includes applying a high-pass filter to

the first audio channel included in the input audio signal to obtain the mid-to-high-frequency components of the first audio channel. The mid-to-high-frequency components of the first audio channel included in the input audio signal include the components of the first audio channel that have frequencies greater than the crossover frequency.

(57) Similarly, virtualized reverberation application **120** further applies respective filtering processes **702** to each additional audio channel included in the input audio signal to separate the low-frequency components of the additional audio channels included in the input audio signal from the mid-to-high-frequency components of the additional audio channels included in the input audio signal. For example, virtualized reverberation application **120** implements Nth filtering process **702-N** to separate the low-frequency components of the Nth audio channel included in the input audio signal from the mid-to-high-frequency components of the Nth audio channel included in the input audio signal.

(58) After separating the low-frequency components of the N audio channels included in the input audio signal from the mid-to-high-frequency components of the N audio channels included in the input audio signal, virtualized reverberation application **120** uses the first processing technique **504**, as described above, to add reverberation to the low-frequency components of the N audio channels included in the input audio signal. Furthermore, virtualized reverberation application **120** uses second and third processing techniques **506**, **508**, as described above, to add reverberation to the mid-to-high-frequency components of the N audio channels included in the input audio signal.

(59) FIG. **8** is a flow chart of method steps for processing an audio signal, according to one or more aspects of the various embodiments. Although the method steps are described with respect to the systems and examples of FIGS. **1-7**, persons skilled in the art will understand that any system configured to perform the method steps, in any order, falls within the scope of the various embodiments.

(60) As shown, a method **800** begins at step **802**, where virtualized reverberation application **120** obtains a measured BRIR of an acoustic space, such as the measured BRIR of acoustic space **202**. In some examples, BRIR measurement application **118** obtains the BRIR by measuring the BRIR of acoustic space **202** and storing the measured BRIR of acoustic space **202** as BRIR data **122** in system storage **112**. In some examples, virtualized reverberation application **120** receives the measured BRIR of acoustic space **202** directly from BRIR measurement application **118**. In such examples, BRIR measurement application **118** measures the BRIR of acoustic space **202** and provides the measured BRIR of acoustic space **202** to virtualized reverberation application **120**. In some examples, BRIR measurement application **118** uses one or more known techniques, such as the MLS technique, the IRS technique, the time-stretched pulses technique, or the logarithmic sinesweep technique, to measure BRIR of acoustic space **202**.

(61) At step **804**, virtualized reverberation application **120** receives an input audio signal. In some examples, the input audio signal includes a single audio channel. In other examples, the input audio signal includes multiple audio channels. In one example, the input audio signal is a stereo audio signal that includes a left audio channel and a right audio channel. In other examples, the input audio signal includes 5.1.2 audio channels, 7.1.2 audio channels, or some other amount of audio channels.

(62) At step **806**, virtualized reverberation application **120** separates the low-frequency components of the input audio signal from the mid-to-high-frequency components of the input audio signal. In some examples, virtualized reverberation application **120** applies a low-pass filter to the input audio signal to obtain low-frequency components of the input audio signal that have frequencies less than a crossover frequency. In such examples, virtualized reverberation application **120** further applies a high-pass filter to the input audio signal to obtain mid-to-high-frequency components of the input audio signal that have frequencies greater than the crossover frequency. In one non-limiting example, the crossover frequency is 900 Hz. In other examples, the crossover frequency can be greater than or less than 900 Hz.

(63) At step **808**, virtualized reverberation application **120** converts the low-frequency components of the input audio signal into a plurality of converted low-frequency audio channels. For example, virtualized reverberation application **120** uses encoding and decoding process **510** to convert (e.g., encode and decode) the low-frequency components of the input audio signal into a plurality of converted low-frequency audio channels. In one non-limiting example, virtualized reverberation application **120** uses a low-order Ambisonics encoder-decoder to encode the low-frequency components of the input audio signal into four low-frequency Ambisonics-encoded audio channels and decode the four low-frequency Ambisonics-encoded audio channels into a plurality of low-frequency decoded audio channels. In other examples, virtualized reverberation application **120** can convert the low-frequency components of the input audio signal using a different type of low-order encoder-decoder.

(64) At step **810**, virtualized reverberation application **120** reduces the respective gains of the plurality of converted low-frequency audio channels. In one non-limiting example, virtualized reverberation application **120** reduces the respective gains of the plurality of converted low-frequency audio channels to a value between -8 dB and -15 dB. In other examples, virtualized reverberation application **120** can reduce the respective gains of the plurality of converted low-frequency audio channels to have different gain values.

(65) At step **812**, virtualized reverberation application **120** convolves the plurality of converted low-frequency audio channels with low-frequency components **400** of the measured BRIR of acoustic space **202**. For example, virtualized reverberation application **120** uses low-frequency convolution process **514** to convolve the plurality of converted low-frequency audio channels with low-frequency components **400** of the measured BRIR of acoustic space **202**. The result of convolving the plurality of converted low-frequency audio channels with the low-frequency components **400** of the measured BRIR is a rendered low-frequency left audio channel that includes reverberation effects and a rendered low-frequency right audio channel that includes reverberation effects.

(66) At step **814**, virtualized reverberation application **120** divides the mid-to-high-frequency components **300** of the measured BRIR of acoustic space **202** into a direct sound portion **302** and a reflected sound portion **304**. In some examples, virtualized reverberation application **120** divides the mid-to-high-frequency components **300** of the measured BRIR of acoustic space **202** into a direct sound portion **302** and a reflected sound portion **304** temporally. In such examples, virtualized reverberation application **120** can divide the mid-to-high-frequency components **300** of the measured BRIR into a first time window **306** that occurs before a time $T1$ and second time window **308** that occurs after the time $T1$ to separate the direct sound portion **302** from the reflected sound portion **304**. In one non-limiting example, the value of time $T1$ is 2 ms. In other examples, the value of time $T1$ can be less than or greater than 2 ms.

(67) At step **816**, virtualized reverberation application **120** converts the mid-to-high-frequency components of the input audio signal into a plurality of converted mid-to-high-frequency audio channels. For example, virtualized reverberation application **120** uses encoding and decoding process **518** to convert (e.g., encode and decode) the mid-to-high-frequency components of the input audio signal into a plurality of converted mid-to-high-frequency audio channels. In one non-limiting example, virtualized reverberation application **120** uses a low-order Ambisonics encoder-decoder to encode the mid-to-high-frequency components of the input audio signal into four mid-to-high-frequency Ambisonics-encoded audio channels and decode the four mid-to-high-frequency Ambisonics-encoded audio channels into a plurality of mid-to-high-frequency decoded audio channels. In other examples, virtualized reverberation application **120** can convert the mid-to-high-frequency components of the input audio signal using a different type of low-order encoder-decoder.

(68) At step **818**, virtualized reverberation application **120** reduces the respective gains of the plurality of converted mid-to-high-frequency audio channels. In one non-limiting example,

virtualized reverberation application **120** reduces the respective gains of the plurality of converted mid-to-high-frequency audio channels to values between -8 dB and -15 dB. In other examples, virtualized reverberation application **120** can reduce the respective gains of the plurality of converted mid-to-high-frequency audio channels to have different gain values.

(69) At step **820**, virtualized reverberation application **120** convolves the plurality of converted mid-to-high-frequency audio channels with the reflected sound portion **304** included in the mid-to-high-frequency components **300** of the measured BRIR of acoustic space **202**. For example, virtualized reverberation application **120** uses reflected convolution process **522** to convolve the plurality of converted mid-to-high-frequency audio channels with the reflected sound portion **304** included in the mid-to-high-frequency components **300** of the measured BRIR of acoustic space **202**. The result of convolving the plurality of converted mid-to-high-frequency audio channels with the reflected sound portion **304** included in the mid-to-high-frequency components **300** of the measured BRIR is a rendered mid-to-high-frequency left audio channel that includes reverberation effects associated with the reflected sound portion of a measured BRIR and a rendered mid-to-high-frequency right car audio channel that includes reverberation effects associated with the reflected sound portion of a measured BRIR.

(70) At step **822**, virtualized reverberation application **120** convolves the discrete mid-to-high-frequency components of the input audio signal with the direct sound portion **302** included in the mid-to-high-frequency components **300** of the measured BRIR of acoustic space **202**. That is, virtualized reverberation application **120** does not convert the mid-to-high-frequency components of the input audio signal before convolving the mid-to-high-frequency components of the input audio signal with the direct sound portion **302** included in the mid-to-high-frequency components **300** of the measured BRIR of acoustic space **202**. For example, virtualized reverberation application **120** uses direct convolution process **526** to convolve the discrete mid-to-high-frequency components of the input audio signal with the direct sound portion **302** included in the mid-to-high-frequency components **300** of the measured BRIR of acoustic space **202**. The result of convolving the discrete mid-to-high-frequency audio channels with the direct sound portion **302** included in the mid-to-high-frequency components **300** of the measured BRIR is a rendered mid-to-high-frequency left audio channel that includes reverberation effects associated with the direct sound portion of a measured BRIR and a rendered mid-to-high-frequency right audio channel that includes reverberation effects associated with the direct sound portion of a measured BRIR.

(71) At step **824**, virtualized reverberation application **120** equalizes the rendered left and right audio channels. For example, virtualized reverberation application **120** applies headphone equalization process **516** to the rendered low-frequency left and right audio channels that include reverberation effects. As another example, virtualized reverberation application **120** applies headphone equalization process **524** to the rendered mid-to-high-frequency left and right audio channels that include reverberation effects associated with the reflected sound portion of a measured BRIR. As another example, virtualized reverberation application **120** applies headphone equalization process **528** to the rendered mid-to-high-frequency left and right audio channels that include reverberation effects associated with the direct sound portion of a measured BRIR.

(72) At step **826**, virtualized reverberation application **120** sums, or combines, the rendered left audio channels and sums, or combines, the rendered right audio channels. For example, virtualized reverberation application **120** implements summation process **530** to combine the rendered low-frequency left audio channel that includes reverberation effects, the rendered mid-to-high-frequency left audio channel that includes reverberation effects associated with the reflected sound portion of a measured BRIR, and the rendered mid-to-high-frequency left audio channel that includes reverberation effects associated with the direct sound portion of a measured BRIR into a single rendered left audio channel that includes reverberation effects. Similarly, virtualized reverberation application **120** implements summation process **530** to combine the rendered low-frequency right audio channel that includes reverberation effects, the rendered mid-to-high-

frequency right audio channel that includes reverberation effects associated with the reflected sound portion of a measured BRIR, and the rendered mid-to-high-frequency right audio channel that includes reverberation effects associated with the direct sound portion of a measured BRIR into a single rendered right audio channel that includes reverberation effects. At step **828**, virtualized reverberation application **120** outputs a rendered stereo signal that includes the single rendered left audio channel that includes reverberation effects and the single rendered right audio channel that includes reverberation effects.

(73) In sum, a computing device renders an audio signal with reverberation based on a measured binaural room impulse response (BRIR) of an acoustic space to add reverberation effects of the acoustic space for playback using headphones. With the disclosed techniques, the computing device adds reverberation to the audio signal using three separate processing techniques. Further, with the disclosed techniques, the computing device separates an audio signal into low-frequency components and high-frequency components. Using a first processing technique, the computing device converts the low-frequency components of the audio signal into a first plurality of converted audio channels and convolves the first plurality of converted audio channels with the measured BRIR to generate a first left audio channel and a first right audio channel. With the disclosed techniques, the computing device further divides the measured BRIR into a first time window that includes the direct sound portion of the measured BRIR and a second time window that includes the reflected sound portion of the measured BRIR. Using a second processing technique, the computing device converts the high-frequency components of the audio signal into a second plurality of converted audio channels and convolves the second plurality of converted audio channels with the reflected sound portion of the measured BRIR to generate a second left audio channel and a second right audio channel. Using a third processing technique, the computing device convolves the discrete high-frequency components of the audio signal with the direct sound portion of the measured BRIR to generate a third left audio channel and a third right audio channel. The direct sound portion of the measured BRIR corresponds to a direction at which sound emitted by a source being rendered arrives directly at the ears of a listener. With the disclosed techniques, the computing device combines the results of the first, second, and third processing techniques to generate left and right car sounds that include reverberation. For example, the computing device combines the first, second, and third left audio channels into a single left audio channel and the combines the first, second, and third right audio channels into a single right audio channel. The computing device then outputs an output audio signal that includes the combined left audio channel and the combined right audio channel for playback by headphones. Before or after combining the left and right car audio channels, the computing device can optionally apply headphone equalization to the left and right audio channels.

(74) At least one technical advantage of the disclosed techniques relative to the prior art is that, with the disclosed techniques, a measured room impulse response of an acoustic space can be used to add reverberation to an audio signal with a lower computational cost. Accordingly, with the disclosed techniques, relatively modest processing power can be used to render reverberation for a large number of sound sources and acoustic spaces based on measured room impulse responses, which sound more natural than room impulse responses derived from models of acoustic spaces. These technical advantages provide one or more technological improvements over prior art approaches.

(75) 1. In some embodiments, a computer-implemented method for audio processing comprises obtaining a binaural room impulse response (BRIR) of an acoustic space; receiving an input audio signal; separating the input audio signal into low-frequency components and high-frequency components; dividing the BRIR of the acoustic space into a first portion that occurs before a first time and a second portion that occurs after the first time; generating a first component of an output audio signal based on the high-frequency components of the input audio signal and the first portion of the BRIR; generating a second component of the output audio signal based on the high-

frequency components of the input audio signal and the second portion of the BRIR; generating a third component of the output audio signal based on the low-frequency components of the input audio signal and the BRIR; and outputting the output audio signal.

(76) 2. The computer-implemented method according to clause 1, wherein the first portion of the BRIR includes a direct sound portion and the second portion of the BRIR includes a reflected sound portion.

(77) 3. The computer-implemented method according to clause 1 or clause 2, wherein generating the first component of the output audio signal comprises convolving the high-frequency components of the input audio signal with the first portion of the BRIR to generate a first left audio channel and a first right audio channel.

(78) 4. The computer-implemented method according to any of clauses 1-3, further comprising applying headphone equalization to the first left audio channel and the first right audio channel.

(79) 5. The computer-implemented method according to any of clauses 1-4, wherein the first portion of the BRIR corresponds to a direction at which sound emitted by a source in the acoustic space arrives directly at an ear of a listener.

(80) 6. The computer-implemented method according to any of clauses 1-5, wherein generating the second component of the output audio signal comprises converting the high-frequency components of the input audio signal into a first plurality of converted audio channels; and convolving the first plurality of converted audio channels with the second portion of the BRIR to generate a second left audio channel and a second right audio channel.

(81) 7. The computer-implemented method according to any of clauses 1-6, wherein converted the high-frequency components of the input audio signal into the first plurality of converted audio channels includes using a first-order Ambisonics encoder-decoder.

(82) 8. The computer-implemented method according to any of clauses 1-7, further comprising reducing a gain of the first plurality of converted audio channels before convolving the first plurality of converted audio channels with the second portion of the BRIR.

(83) 9. The computer-implemented method according to any of clauses 1-8, wherein generating the third component of the output audio signal comprises converting the low-frequency components of the input audio signal into a second plurality of converted audio channels; and convolving the second plurality of converted audio channels with the BRIR to generate a third left audio channel and a third right audio channel.

(84) 10. The computer-implemented method according to any of clauses 1-9, wherein obtaining the BRIR of the acoustic space includes measuring the BRIR of the acoustic space with a microphone.

(85) 11. The computer-implemented method according to any of clauses 1-10, wherein the output audio signal comprises a left channel and a right channel.

(86) 12. According to some embodiments, one or more non-transitory computer-readable storage media including instructions that, when executed by one or more processors at a computing device, cause the one or more processors to perform steps of obtaining a binaural room impulse response (BRIR) of an acoustic space; receiving an input audio signal; separating the input audio signal into low-frequency components and high-frequency components; dividing the BRIR of the acoustic space into a first portion that occurs before a first time and a second portion that occurs after the first time; generating a first component of an output audio signal based on the high-frequency components of the input audio signal and the first portion of the BRIR; generating a second component of the output audio signal based on the high-frequency components of the input audio signal and the second portion of the BRIR; generating a third component of the output audio signal based on the low-frequency components of the input audio signal and the BRIR; and outputting the output audio signal.

(87) 13. The one or more non-transitory computer-readable media according to clause 12, wherein the first portion of the BRIR includes a direct sound portion and the second portion of the BRIR includes a reflected sound portion.

(88) 14. The one or more non-transitory computer-readable media according to clause 12 or clause 13, wherein generating the first component of the output audio signal comprises convolving the high-frequency components of the input audio signal with the first portion of the BRIR to generate a first left audio channel and a first right audio channel; generating the second component of the output audio signal comprises converting the high-frequency components of the input audio signal into a first plurality of converted audio channels and convolving the first plurality of converted audio channels with the second portion of the BRIR to generate a second left audio channel and a second right audio channel; and generating the third component of the output audio signal comprises converting the low-frequency components of the input audio signal into a second plurality of converted audio channels and convolving the second plurality of converted audio channels with the BRIR to generate a third left audio channel and a third right audio channel.

(89) 15. The one or more non-transitory computer-readable media according to any of clauses 12-14, wherein the steps further comprise applying headphone equalization to the first left audio channel and the first right audio channel; applying headphone equalization to the second left audio channel and the second right audio channel; and applying headphone equalization to the third left audio channel and the third right audio channel.

(90) 16. The one or more non-transitory computer-readable media according to any of clauses 12-15, wherein the steps further comprise reducing a gain of the first plurality of converted audio channels before convolving the first plurality of converted audio channels with the second portion of the BRIR; and reducing a gain of the second plurality of converted audio channels before convolving the second plurality of converted audio channels with the BRIR.

(91) 17. The one or more non-transitory computer-readable media according to any of clauses 12-16, wherein converting the high-frequency components of the input audio signal into the first plurality of converted audio channels includes using a first-order Ambisonics encoder-decoder; and converting the low-frequency components of the input audio signal into the second plurality of converted audio channels includes using a first-order Ambisonics encoder-decoder.

(92) 18. The one or more non-transitory computer-readable media according to any of clauses 12-17, wherein obtaining the BRIR of the acoustic space includes measuring the BRIR of the acoustic space with a microphone.

(93) 19. According to some embodiments, a computing device comprising a memory storing an application and one or more processors that, when executing the application, are configured to obtain a binaural room impulse response (BRIR) of an acoustic space; receive an input audio signal; separate the input audio signal into low-frequency components and high-frequency components; divide the BRIR of the acoustic space into a first portion that occurs before a first time and a second portion that occurs after the first time; generate a first component of an output audio signal based on the high-frequency components of the input audio signal and the first portion of the BRIR; generate a second component of the output audio signal based on the high-frequency components of the input audio signal and the second portion of the BRIR; generate a third component of the output audio signal based on the low-frequency components of the input audio signal and the BRIR; and output the output audio signal.

(94) 20. The computing device according to clause 19, wherein the first portion of the BRIR includes a direct sound portion and the second portion of the BRIR includes a reflected sound portion.

(95) Any and all combinations of any of the claim elements recited in any of the claims and/or any elements described in this application, in any fashion, fall within the contemplated scope of the present disclosure and protection.

(96) The descriptions of the various embodiments have been presented for purposes of illustration, but are not intended to be exhaustive or limited to the embodiments disclosed. Many modifications and variations will be apparent to those of ordinary skill in the art without departing from the scope and spirit of the described embodiments.

(97) Aspects of the present embodiments may be embodied as a system, method or computer program product. Accordingly, aspects of the present disclosure may take the form of an entirely hardware embodiment, an entirely software embodiment (including firmware, resident software, micro-code, etc.) or an embodiment combining software and hardware aspects that may all generally be referred to herein as a “module,” a “system,” or a “computer.” In addition, any hardware and/or software technique, process, function, component, engine, module, or system described in the present disclosure may be implemented as a circuit or set of circuits. Furthermore, aspects of the present disclosure may take the form of a computer program product embodied in one or more computer readable medium(s) having computer readable program code embodied thereon.

(98) Any combination of one or more computer readable medium(s) may be utilized. The computer readable medium may be a computer readable signal medium or a computer readable storage medium. A computer readable storage medium may be, for example, but not limited to, an electronic, magnetic, optical, electromagnetic, infrared, or semiconductor system, apparatus, or device, or any suitable combination of the foregoing. More specific examples (a non-exhaustive list) of the computer readable storage medium would include the following: an electrical connection having one or more wires, a portable computer diskette, a hard disk, a random access memory (RAM), a read-only memory (ROM), an erasable programmable read-only memory (EPROM or Flash memory), an optical fiber, a portable compact disc read-only memory (CD-ROM), an optical storage device, a magnetic storage device, or any suitable combination of the foregoing. In the context of this document, a computer readable storage medium may be any tangible medium that can contain, or store a program for use by or in connection with an instruction execution system, apparatus, or device.

(99) Aspects of the present disclosure are described above with reference to flowchart illustrations and/or block diagrams of methods, apparatus (systems) and computer program products according to embodiments of the disclosure. It will be understood that each block of the flowchart illustrations and/or block diagrams, and combinations of blocks in the flowchart illustrations and/or block diagrams, can be implemented by computer program instructions. These computer program instructions may be provided to a processor of a general purpose computer, special purpose computer, or other programmable data processing apparatus to produce a machine. The instructions, when executed via the processor of the computer or other programmable data processing apparatus, enable the implementation of the functions/acts specified in the flowchart and/or block diagram block or blocks. Such processors may be, without limitation, general purpose processors, special-purpose processors, application-specific processors, or field-programmable gate arrays.

(100) The flowchart and block diagrams in the figures illustrate the architecture, functionality, and operation of possible implementations of systems, methods and computer program products according to various embodiments of the present disclosure. In this regard, each block in the flowchart or block diagrams may represent a module, segment, or portion of code, which comprises one or more executable instructions for implementing the specified logical function(s). It should also be noted that, in some alternative implementations, the functions noted in the block may occur out of the order noted in the figures. For example, two blocks shown in succession may, in fact, be executed substantially concurrently, or the blocks may sometimes be executed in the reverse order, depending upon the functionality involved. It will also be noted that each block of the block diagrams and/or flowchart illustration, and combinations of blocks in the block diagrams and/or flowchart illustration, can be implemented by special purpose hardware-based systems that perform the specified functions or acts, or combinations of special purpose hardware and computer instructions.

(101) While the preceding is directed to embodiments of the present disclosure, other and further

embodiments of the disclosure may be devised without departing from the basic scope thereof, and the scope thereof is determined by the claims that follow.

Claims

1. A computer-implemented method for audio processing, the method comprising: obtaining a binaural room impulse response (BRIR) of an acoustic space; receiving an input audio signal; separating the input audio signal into low-frequency components and high-frequency components; dividing the BRIR of the acoustic space into a first portion that occurs before a first time and a second portion that occurs after the first time; generating a first component of an output audio signal based on the high-frequency components of the input audio signal and the first portion of the BRIR; generating a second component of the output audio signal based on the high-frequency components of the input audio signal and the second portion of the BRIR; generating a third component of the output audio signal based on the low-frequency components of the input audio signal and the BRIR; and outputting the output audio signal.
2. The computer-implemented method of claim 1, wherein the first portion of the BRIR includes a direct sound portion and the second portion of the BRIR includes a reflected sound portion.
3. The computer-implemented method of claim 1, wherein generating the first component of the output audio signal comprises convolving the high-frequency components of the input audio signal with the first portion of the BRIR to generate a first left audio channel and a first right audio channel.
4. The computer-implemented method of claim 3, further comprising applying headphone equalization to the first left audio channel and the first right audio channel.
5. The computer-implemented method of claim 1, wherein the first portion of the BRIR corresponds to a direction at which sound emitted by a source in the acoustic space arrives directly at an ear of a listener.
6. The computer-implemented method of claim 1, wherein generating the second component of the output audio signal comprises: converting the high-frequency components of the input audio signal into a first plurality of converted audio channels; and convolving the first plurality of converted audio channels with the second portion of the BRIR to generate a second left audio channel and a second right audio channel.
7. The computer-implemented method of claim 6, wherein converted the high-frequency components of the input audio signal into the first plurality of converted audio channels includes using a first-order Ambisonics encoder-decoder.
8. The computer-implemented method of claim 6, further comprising reducing a gain of the first plurality of converted audio channels before convolving the first plurality of converted audio channels with the second portion of the BRIR.
9. The computer-implemented method of claim 1, wherein generating the third component of the output audio signal comprises: converting the low-frequency components of the input audio signal into a second plurality of converted audio channels; and convolving the second plurality of converted audio channels with the BRIR to generate a third left audio channel and a third right audio channel.
10. The computer-implemented method of claim 1, wherein obtaining the BRIR of the acoustic space includes measuring the BRIR of the acoustic space with a microphone.
11. The computer-implemented method of claim 1, wherein the output audio signal comprises a left channel and a right channel.
12. One or more non-transitory computer-readable storage media including instructions that, when executed by one or more processors at a computing device, cause the one or more processors to perform steps of: obtaining a binaural room impulse response (BRIR) of an acoustic space; receiving an input audio signal; separating the input audio signal into low-frequency components

and high-frequency components; dividing the BRIR of the acoustic space into a first portion that occurs before a first time and a second portion that occurs after the first time; generating a first component of an output audio signal based on the high-frequency components of the input audio signal and the first portion of the BRIR; generating a second component of the output audio signal based on the high-frequency components of the input audio signal and the second portion of the BRIR; generating a third component of the output audio signal based on the low-frequency components of the input audio signal and the BRIR; and outputting the output audio signal.

13. The one or more non-transitory computer-readable storage media of claim 12, wherein the first portion of the BRIR includes a direct sound portion and the second portion of the BRIR includes a reflected sound portion.

14. The one or more non-transitory computer-readable storage media of claim 12, wherein: generating the first component of the output audio signal comprises convolving the high-frequency components of the input audio signal with the first portion of the BRIR to generate a first left audio channel and a first right audio channel; generating the second component of the output audio signal comprises: converting the high-frequency components of the input audio signal into a first plurality of converted audio channels; and convolving the first plurality of converted audio channels with the second portion of the BRIR to generate a second left audio channel and a second right audio channel; and generating the third component of the output audio signal comprises: converting the low-frequency components of the input audio signal into a second plurality of converted audio channels; and convolving the second plurality of converted audio channels with the BRIR to generate a third left audio channel and a third right audio channel.

15. The one or more non-transitory computer-readable storage media of claim 14, wherein the steps further comprise: applying headphone equalization to the first left audio channel and the first right audio channel; applying headphone equalization to the second left audio channel and the second right audio channel; and applying headphone equalization to the third left audio channel and the third right audio channel.

16. The one or more non-transitory computer-readable storage media of claim 14, wherein the steps further comprise: reducing a gain of the first plurality of converted audio channels before convolving the first plurality of converted audio channels with the second portion of the BRIR; and reducing a gain of the second plurality of converted audio channels before convolving the second plurality of converted audio channels with the BRIR.

17. The one or more non-transitory computer-readable storage media of claim 14, wherein: converting the high-frequency components of the input audio signal into the first plurality of converted audio channels includes using a first-order Ambisonics encoder-decoder; and converting the low-frequency components of the input audio signal into the second plurality of converted audio channels includes using a first-order Ambisonics encoder-decoder.

18. The one or more non-transitory computer-readable storage media of claim 12, wherein obtaining the BRIR of the acoustic space includes measuring the BRIR of the acoustic space with a microphone.

19. A computing device comprising: a memory storing an application; and one or more processors that, when executing the application, are configured to: obtain a binaural room impulse response (BRIR) of an acoustic space; receive an input audio signal; separate the input audio signal into low-frequency components and high-frequency components; divide the BRIR of the acoustic space into a first portion that occurs before a first time and a second portion that occurs after the first time; generate a first component of an output audio signal based on the high-frequency components of the input audio signal and the first portion of the BRIR; generate a second component of the output audio signal based on the high-frequency components of the input audio signal and the second portion of the BRIR; generate a third component of the output audio signal based on the low-frequency components of the input audio signal and the BRIR; and output the output audio signal.

20. The computing device of claim 19, wherein the first portion of the BRIR includes a direct sound portion and the second portion of the BRIR includes a reflected sound portion.
